

Problem Set 3: Instrumental Variables

MY457 | WT 2026 | Week 7

1 Concepts

- 1.1 In the table below, fill in the missing values and **bold** the values you fill in. In one case, this is not possible. Which one is it and why? Note that Y_{1i} and Y_{0i} are the potential outcomes with respect to encouragement Z , so that $Y_i = Y_{Z_i,i}$.

Table 1: Table of Potential Outcomes

i	Z_i	D_{0i}	D_{1i}	D_i	Compliance Type	Y_{0i}	Y_{1i}	Y_i
1	0				Never-taker	3	3	
2	1	0	1			2	5	
3	1	1	1			4	4	
4	0	1				3	0	3
5	0	1		1				4

Answer

i	Z_i	D_{0i}	D_{1i}	D_i	Compliance Type	Y_{0i}	Y_{1i}	Y_i
1	0	0	0	0	Never-taker	3	3	3
2	1	0	1	1	Complier	2	5	5
3	1	1	1	1	Always-taker	4	4	4
4	0	1	0	1	Defier	3	0	3
5	0	1		1		4		4

Note: For unit 1 (never-taker), we fill in $Y_{1i} = Y_{0i} = 3$. Since the never-taker does not take treatment regardless of encouragement ($D_{0i} = D_{1i} = 0$), the exclusion restriction implies that their outcome is the same whether or not they are encouraged.

For unit 2 (complier), $Z_2 = 1$ so $D_2 = D_{12} = 1$ and $Y_2 = Y_{1i} = 5$.

For unit 3 (always-taker), $Z_3 = 1$ so $D_3 = D_{13} = 1$ and $Y_3 = Y_{1i} = 4$.

For unit 4 (defier), $Z_4 = 0$ so $D_4 = D_{04} = 1$ and $Y_4 = Y_{0i} = 3$.

For unit 5, $Z_5 = 0$, so $Y_5 = Y_{05} = 4$, so we can fill in Y_{05} . However, we cannot determine D_{15} : subject $i = 5$ could be either an always-taker ($D_{15} = 1$) or a defier ($D_{15} = 0$). We didn't encourage unit 5 to take the treatment, but they took it anyway. This could be because 5 always takes the treatment, regardless of encouragement, or because 5 is a defier who only takes the treatment when not encouraged. Without knowing D_{15} , we also cannot determine Y_{15} .

- 1.2 Write down the estimand for the ITT. Excluding the unit for which you can't fill in the missing values, calculate the ITT using the potential outcomes in Table 1.

Answer

The ITT is defined as the average difference between the potential outcomes under encouragement and no encouragement. You can write it as an expected value of the difference of potential outcomes under encouragement and no encouragement:

$$\tau_{ITT} = E[Y_{(1,D_{1i})i} - Y_{(0,D_{0i})i}]$$

or as the average of the observed differences in outcomes between the encouraged and non-encouraged groups:

$$\tau_{ITT} = \frac{1}{N} \sum_{i=1}^N (Y_{1i} - Y_{0i})$$

plugging in the values from units $i \in \{1, 2, 3, 4\}$ Table 1, we can calculate the ITT as follows:

$$\begin{aligned}
\tau_{ITT} &= \frac{1}{4} \sum_{i=1}^4 (Y_{1i} - Y_{0i}) \\
&= \frac{1}{4} ((3 - 3) + (5 - 2) + (4 - 4) + (0 - 3)) \\
&= \frac{1}{4} (0 + 3 + 0 - 3) \\
&= \frac{0}{4} \\
&= 0
\end{aligned}$$

1.3 In this example, is the ITT equal to the ATE (where the ATE is defined with respect to treatment D)? Why or why not?

 Answer

While we can't determine the ATE from the data in Table 1, there clearly is non-compliance in the data, which means that the ITT will not be equal to the ATE.

1.4 Beyond SUTVA, what must we assume to identify the LATE? From the data in Table 1, can you determine whether these assumptions hold? Why or why not?

 Answer

The assumptions we need to make to identify the LATE are:

1. *Relevance*: The instrument has an effect on treatment status. This implies that there are some compliers in the data. We can see from Table 1 that there is at least one complier (unit 2), so the relevance assumption holds. However, we can't really assess the strength of the instrument in any meaningful way with such a small sample size.
2. *Ignorability* or *exogeneity*: The potential outcome of the outcome and the potential outcome of the treatment are independent of the instrument. This means that there are no confounders $Z \leftarrow U \rightarrow Y$ and that there are no confounders $Z \leftarrow U \rightarrow D$, where U are the confounders. Formally: $\{Y_{ZD}, D_Z\} \perp\!\!\!\perp Z$. We can't determine whether this assumption holds from the data in Table 1 because we don't observe the data generating process.
3. *Exclusion Restriction*: The instrument only affects the outcome through its effect

on treatment. $Y_{1,d} = Y_{0,d}$ for all d . This implies that there are no direct effects of the instrument on the outcome and that there are no alternative pathways from the instrument to the outcome that don't go through the treatment. In Table 1, we can see that the exclusion restriction holds for the four units for which we can fill in the missing values. This is because Z only affects the potential outcomes through D . Among always-takers or never-takers, the potential outcomes with respect to the encouragement are the same under encouragement and no encouragement. Encouragement doesn't affect the potential outcomes, unless it changes treatment status.

4. *Monotonicity*: There are no defiers. This means that $D_{1i} \geq D_{0i}$ for all i ; encouragement weakly increases treatment uptake. We can see from Table 1 that there is at least one defier (unit 4), so the monotonicity assumption does not hold.

2 Simulation

- 2.1 Explain the code below and relate it to an instrumental variables data generating process. Calculate the true ITT and ATE.

```
set.seed(123)
n_obs <- 1000

data <- tibble(
  U = rbinom(n_obs, 1, .75),
  c_type = ifelse(U == 1,
    sample(1:3, n_obs, prob = c(0.7, 0.1, 0.2), replace = T),
    sample(1:3, n_obs, prob = c(0.35, 0.3, 0.35), replace = T)
  ),
  tau = case_match(c_type, 1 ~ 5000, 2 ~ 1000, 3 ~ 2500),
  Z = rbinom(n_obs, 1, .5),
  D = case_match(c_type, 1 ~ as.integer(Z == 1), 2 ~ 1L, 3 ~ 0L),
  Y0 = rnorm(n_obs, mean = 50000, sd = 2500) + 25000 * U,
  Y1 = Y0 + tau,
  Y = ifelse(D == 1, Y1, Y0)
)
```

Answer

The code above simulates data for an instrumental variables setting. We simulate 1000 observations and the DGP includes the following variables:

Variable	Description
U	Unobserved variable that affects both compliance type and potential outcomes.
c_type	Compliance type, which can be one of three types: 1 (complier), 2 (always-taker), or 3 (never-taker). The probability of being each type depends on the value of U. There are no defiers.
tau	The treatment effect, which depends on the compliance type. Compliers have a treatment effect of 5000, always-takers have a treatment effect of 1000, and never-takers have a treatment effect of 2500.
Z	The instrument, which is randomly assigned with a 50% chance of being 1 (encouraged) or 0 (not encouraged).
D	The treatment status, which depends on the compliance type and the instrument.
Y0	The potential outcome under control, which is normally distributed with a mean of 50000 and a standard deviation of 2500, plus an additional 25000 if U is 1.
Y1	The potential outcome under treatment, which is equal to Y0 plus the treatment effect tau for the specific unit.
Y	The observed outcome, which is equal to Y1 if the unit receives treatment (D=1) and Y0 if the unit does not receive treatment (D=0).

Because we observe both potential outcomes with respect to the treatment, we can simply calculate the true ATE as the average difference between Y_1 and Y_0 across all units:

```
true_ate <- mean(data$Y1 - data$Y0)
true_ate
```

```
[1] 3789.5
```

```
# also correct:
mean(data$tau)
```

```
[1] 3789.5
```

Note that this is equivalent to a weighted average of the treatment effects for each compliance type, where the weights are the proportions of each compliance type in the population:

```
true_ate_decomposed <- {
  mean(data$c_type == 1) * mean(data$tau[data$c_type == 1]) +
  mean(data$c_type == 2) * mean(data$tau[data$c_type == 2]) +
  mean(data$c_type == 3) * mean(data$tau[data$c_type == 3])
}

true_ate_decomposed == true_ate
```

```
[1] TRUE
```

The ITT is the average difference in potential outcomes with respect to the *instrument* between the encouraged and non-encouraged groups. To calculate it, we first need to calculate the potential outcomes with respect to the instrument.

```
data <- data |> mutate(
  # compliers & always-takers take treatment
  Y_z1 = ifelse(c_type %in% c(1, 2), Y1, Y0),
  # only always-takers take treatment
  Y_z0 = ifelse(c_type == 2, Y1, Y0)
)

true_itt <- mean(data$Y_z1 - data$Y_z0)
true_itt
```

```
[1] 3035
```

Again, you could also estimate the ITT as a weighted average of the ITTs for each compliance type, where the weights are the proportions of each compliance type in the population.

```
prop_complier <- mean(data$c_type == 1)
prop_always <- mean(data$c_type == 2)
prop_never <- mean(data$c_type == 3)
itt_complier <- {
  mean(data$Y_z1[data$c_type == 1]) -
  mean(data$Y_z0[data$c_type == 1])
}
itt_always <- {
  mean(data$Y_z1[data$c_type == 2]) -
  mean(data$Y_z0[data$c_type == 2])
}
itt_never <- {
  mean(data$Y_z1[data$c_type == 3]) -
  mean(data$Y_z0[data$c_type == 3])
}

true_itt_decomposed <- (
  prop_complier * itt_complier
  + prop_always * itt_always
  + prop_never * itt_never
)
```

As the exclusion restriction holds in this DGP, the ITT for always-takers and never-takers is zero, so the ITT is simply the proportion of compliers times the treatment effect for compliers:

```
true_itt_simple <- mean(data$c_type == 1) * mean(data$tau[data$c_type == 1])
identical(true_itt_simple, true_itt) && identical(true_itt, true_itt_decomposed)
```

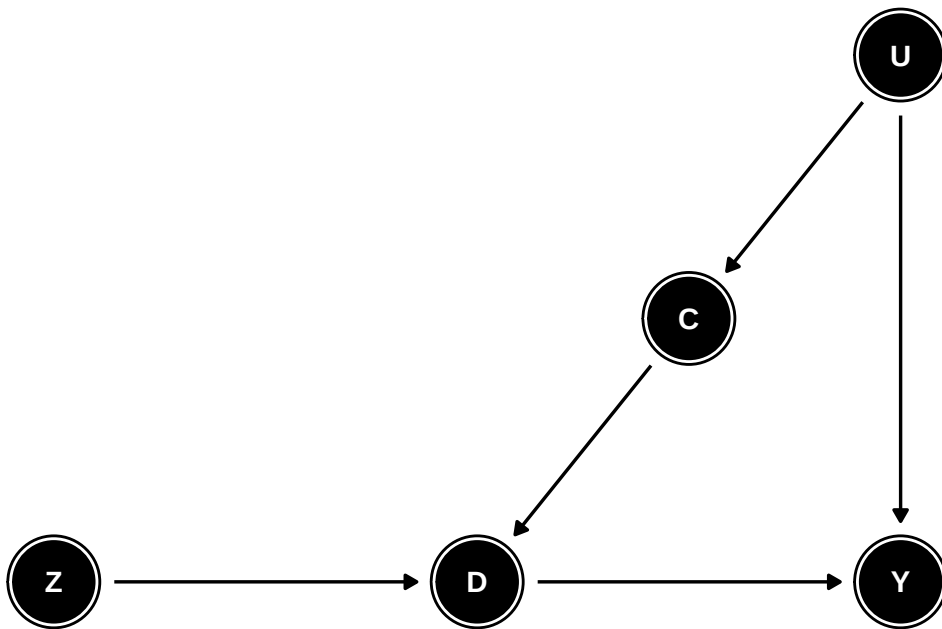
```
[1] TRUE
```

2.2 Create a DAG that represents the data generating process in the code above. Include the variables Z , C (for `c_type`), D , Y , and U .

💡 Answer

The DAG for the data generating process can be represented as follows:

```
iv_dag <- dagify(  
  C ~ U,  
  D ~ C + Z,  
  Y ~ D + U,  
  exposure = "D",  
  outcome = "Y",  
  latent = c("U", "C"),  
  coords = list(  
    x = c(Z = 1, C = 2.5, D = 2, Y = 3, U = 3),  
    y = c(Z = 0, C = .5, D = 0, Y = 0, U = 1)  
  )  
)  
  
ggdag(iv_dag, stylized = T) +  
  theme_dag()
```



2.3 Use OLS to estimate the difference-in-means with respect to treatment D . Is this an unbiased estimator for the ATE in this setting? Why or why not?

💡 Answer

To estimate the difference-in-means with respect to treatment D , we can run the following OLS regression:

```
ols_diff_in_means <- lm(Y ~ D, data = data)
modelsummary(ols_diff_in_means,
  statistic = "p.value",
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)
```

	(1)
(Intercept)	69572.820 (<0.001)
D	1791.252 (0.014)
Num.Obs.	1000
R2	0.006

Recall that the difference-in-means is an unbiased estimator for the ATE if potential outcomes are independent of treatment assignment. As you can see from the DAG above, there is a backdoor path from D to Y through U and C : $D \leftarrow C \leftarrow U \rightarrow Y$. We can't control for this because both U and C are unobserved. Hence, the difference-in-means estimator doesn't estimate the ATE in this setting. In other words, because compliance and potential outcomes are both affected by the unobserved variable U , the relationship between D and Y is confounded and the difference-in-means estimator is biased for the ATE.

2.4 Now estimate the difference-in-means with respect to encouragement Z . Is this an unbiased estimator for any causal effect in this setting? Why or why not?

💡 Answer

To estimate the difference-in-means with respect to encouragement Z , we can run the following OLS regression:

```
ols_diff_in_means_z <- lm(Y ~ Z, data = data)
modelsummary(ols_diff_in_means_z,
  statistic = "p.value",
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)
```

	(1)
(Intercept)	68932.564 (<0.001)
Z	2844.179 (<0.001)
Num.Obs.	1000
R2	0.015

The relationship between Z and Y is not confounded because there are no backdoor paths from Z to Y . The difference-in-means estimator for Z is an unbiased estimator for the ITT in this setting. Substantively, encouragement increased the outcome by about 1427 compared to no encouragement.

2.5 Estimate the LATE using the Wald estimator.

💡 Answer

Recall that the LATE is the average treatment effect among compliers. Under the exclusion restriction and monotonicity, the ITT equals the LATE multiplied by the share of compliers. Hence, if we divide the ITT by the share of compliers, we recover the LATE:

$$\tau_{LATE} = \frac{\tau_{ITT}}{Pr(\text{compliers})}$$

Note that the denominator — the share of compliers — is estimated by the difference

in treatment rates between the encouraged and non-encouraged groups ($E[D_i|Z_i = 1] - E[D_i|Z_i = 0]$). Hence the Wald estimator can be written as:

$$\tau_{LATE} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]}$$

which is also equivalent to the ratio of the covariance of Y and Z to the covariance of D and Z :

$$\tau_{LATE} = \frac{Cov(Y, Z)}{Cov(D, Z)}$$

To estimate the LATE using the Wald estimator, we can run the following code:

```
LATE <- cov(data$Y, data$Z) / cov(data$D, data$Z)
```

- 2.6 Using the Two Stage Least Squares (2SLS) estimator, re-estimate the LATE. Do this both manually (using two `lm()` commands) and using the `AER::ivreg` command. How does your result compare to the previous result? Make sure to present the results of your second stage in a well-formatted table that includes standard errors. Interpret the results and potential differences between the two models.

Answer

To estimate the LATE using the Two Stage Least Squares (2SLS) estimator manually, we can run the following code:

```
# First stage: regress D on Z
first_stage <- lm(D ~ Z, data = data)
data$D_hat <- predict(first_stage)
# Second stage: regress Y on D_hat
second_stage <- lm(Y ~ D_hat, data = data)
```

To estimate the LATE using the `AER::ivreg` command, we can run the following code:

```
library(AER)
iv_model <- ivreg(Y ~ D | Z, data = data)
```

Finally, we can present the results of the second stage in a well-formatted table that includes standard errors:

```

modelsummary(
  list(
    "2SLS (manual)" = second_stage,
    "2SLS (ivreg)" = iv_model
  ),
  coef_map = c("D_hat" = "D_hat", "D" = "D_hat"),
  statistic = "std.error",
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)

```

	2SLS (manual)	2SLS (ivreg)
D_hat	4698.147 (1192.682)	4698.147 (1207.796)
Num.Obs.	1000	1000
R2	0.015	-0.010

Both approaches yield the same point estimate, which also matches the Wald estimator. The key difference is in the standard errors: the manual 2SLS standard errors are **incorrect** because the second-stage `lm()` treats $\hat{D}$ as if it were observed data, ignoring estimation uncertainty from the first stage. The `ivreg` function properly accounts for this and produces correct (larger) standard errors.

Note that you can also create tables that include the first stage results alongside the second stage results. This is very helpful to establish the relevance of the instrument (via the F-statistic). Depending on which package you use, the most efficient way to create such a table may differ.

```
modelsummary(
  list(
    "First Stage" = first_stage,
    "Second Stage (manual)" = second_stage,
    "Second Stage (ivreg)" = iv_model
  ),
  coef_map = c("Z" = "Z", "D_hat" = "D_hat", "D" = "D_hat"),
  statistic = "std.error",
  gof_map = c("nobs", "r.squared", "F"),
  output = "markdown"
)
```

	First Stage	Second Stage (manual)	Second Stage (ivreg)
Z	0.605 (0.025)		
D_hat		4698.147 (1192.682)	4698.147 (1207.796)
Num.Obs.	1000	1000	1000
R2	0.371	0.015	-0.010
F	588.042	15.517	

3 Replication

In this section, we will use real-world data to reinforce what we have learned. We will analyse the dataset employed in *Foreign Aid, Human Rights and Democracy Promotion: Evidence from a Natural Experiment*.

Many countries, especially lower- and middle-income countries, are provided foreign aid with the intention that this will improve the living conditions of those in need. This foreign aid might

have some spillover effects by encouraging the protection of human rights and entrenching democratic institutions. Does foreign aid improve human rights and democracy?

The authors use instrumental variables to study the effect of foreign aid, measured as overseas development assistance (`aid`) on the the CIRI Empowerment Index (`CIRI`).

3.1 Explain in your own words the instrumental variable (IV) design that the authors use to answer the research question. What are the instrument, the treatment, and the outcome?

 Answer

This study aims to determine whether increased foreign aid improves human rights and democracy. Due to the numerous confounding variables that influence both the receipt of foreign aid and the state of human rights and democracy in a country, a simple comparison is not feasible. To address this, the authors identify an “as-if random” instrument to eliminate confounders: whether a country’s former colonizer holds the presidency of the Council of the European Union. The presidency of the Council rotates among member countries in an “essentially random” manner, according to the authors. They demonstrate the instrument’s relevance by showing a positive association between a country’s former colonizer holding the presidency and the level of EU aid the country receives.

Z: Whether a country’s former colonizer holds the presidency of the Council of the European Union. More specifically, the twice lagged former Colony Status in the second half of the year, when budget decisions are made (instrument). *D*: The amount of overseas development assistance (`aid`) a country receives (treatment). *Y*: The CIRI Human Empowerment Index score averaged over 4 years (outcome).

3.2 Give one example for a potential exclusion restriction violation in this setting. How plausible do you think the exclusion restriction is in this case?

 Answer

One potential violation of the exclusion restriction in this setting could be that the presidency of the Council of the European Union has a direct effect on the CIRI Empowerment Index that is not mediated through foreign aid. For example, the presidency might lead to increased diplomatic engagement or other forms of support that could directly influence human rights and democracy in the former colony, independent of the amount of aid received.

3.3 What would have to be true for the ignorability assumption to be violated in this setting? Do you think the ignorability assumption holds?

 Answer

If there are confounders that affect both the likelihood of a country's former colonizer holding the presidency of the Council of the European Union and the CIRI Empowerment Index, or if there are confounders that affect both the likelihood of a country's former colonizer holding the presidency and the amount of aid received, then the ignorability assumption would be violated.

3.4 Read into R the replication data set (`foreign_aid.dta`) and assign it to an object called `aid_data`. Each row in the data is a country-year observation. Estimate the ITT on the CIRI Empowerment Index (CIRI), controlling for country and year fixed effects. Be sure to present your results neatly, showing only relevant statistics for `Colony` and not for all your fixed effects. What do you find?

 Answer

```
aid_data <- read_dta(here("2026", "iv", "data", "foreign_aid.dta"))
aid_data <- aid_data |>
  mutate(
    ccode = as.factor(ccode),
    year = as.factor(year)
  )

itt_mod <- lm(CIRI ~ Colony + factor(year) + factor(ccode), data = aid_data)
modelsummary(list("CIRI" = itt_mod),
  statistic = "p.value",
  coef_map = c("Colony" = "Colony (ITT)"),
  gof_map = c("nobs", "r.squared"),
  add_rows = tribble(
    ~term, ~CIRI,
    "Country FE", "Yes",
    "Year FE", "Yes"
  ),
  output = "markdown",
  stars = T
)
```

	CIRI
Colony (ITT)	0.302* (0.018)
Num.Obs.	1792
R2	0.896
Country FE	Yes
Year FE	Yes

If we only include `Colony` and the fixed effects, we estimate an ITT of 0.3, which is significant at $\alpha = 0.05$. Hence, when a country's former colonizer holds the presidency of the Council of the European Union, the CIRI Empowerment Index of the former colony on average increases by 0.3 points.

- 3.5 Test whether the instrument is relevant. Use the `eff_F()` function from the `ivDiag` package. Your model should instrument `Aid` (`aid`) with presidency status (`Colony`), include country and year fixed effects as controls, and cluster standard errors by country and year. As a rule of thumb, an F-statistic greater than 10 is often considered to indicate a relevant instrument ([Andrews, Stock, and Sun, 2019](#)). What do you find?

Answer

```
F_stat <- ivDiag::eff_F(
  Y = "CIRI",
  D = "aid",
  Z = "Colony",
  FE = c("ccode", "year"), # covariates including FEs
  data = aid_data,
  cl = c("ccode", "year") # multi-way clustering
)
```

The effective F-statistic is 11.38, which is slightly larger than 10, which indicates relevance. However, there are several other robustness checks that the model doesn't pass. If you'd like to dig deeper, check out the output of the following code and read this [paper](#).

```
robustness <- ivDiag(
  Y = "CIRI",
  D = "aid",
  Z = "Colony",
  FE = c("ccode", "year"), # covariates including FEs
  data = aid_data,
  cl = c("ccode", "year") # multi-way clustering
)
```

3.6 Estimate the first stage and the second stage, again controlling for two-way fixed effects. Again, use `lm()` and present the results neatly. What do you find?

 Answer

```
# First stage
reg1 <- lm(
  aid ~ Colony + factor(year) + factor(ccode),
  data = aid_data
)

# Second stage
reg2 <- lm(
  CIRI ~ reg1$fitted.values + factor(year) + factor(ccode),
  data = aid_data
)

modelsummary(
  list(
    "First Stage" = reg1,
    "Second Stage" = reg2
  ),
  coef_map = c("Colony" = "Colony", "reg1$fitted.values" = "D_hat"),
  statistic = "std.error",
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)
```

	First Stage	Second Stage
Colony	0.160	

		(0.073)	
D_hat			1.885
			(0.797)
Num.Obs.	1792		1792
R2	0.764		0.896

The second stage shows a treatment effect of 1.88 which is significant at $\alpha = 0.05$. These results are consistent with the findings of the authors.

3.7 Using the `AER` package, use the `ivreg` function and estimate the LATE. Interpret the LATE substantively, are your results the same as above? Why?

 Answer

```
ivreg(CIRI ~ aid + factor(year) + factor(ccode) | factor(year) + factor(ccode) + Colony, da
  modelsummary(
    statistic = "std.error",
    coef_map = c("aid" = "Effect of Aid (LATE)"),
    gof_map = c("nobs", "r.squared"),
    output = "markdown",
    add_rows = tribble(
      ~term, ~CIRI,
      "Country FE", "Yes",
      "Year FE", "Yes"
    )
  )
)
```

	(1)
Effect of Aid (LATE)	1.885 (1.096)
Num.Obs.	1792
R2	0.804
Country FE	Yes
Year FE	Yes

We get the same point estimate as with the manual 2SLS approach, however, the standard errors change. The manual 2SLS standard errors are incorrect because they treat the fitted values $\hat{D}$ as if they were observed data, ignoring the estimation uncertainty from the first stage. The `ivreg` function properly accounts for this first-stage estimation error when computing standard errors.

- 3.8 Estimate the LATE using the plug-in estimator. What do you find? Are your results different from before? Why?

💡 Answer

```
cov(aid_data$CIRI, aid_data$Colony) / cov(aid_data$aid, aid_data$Colony)
```

```
[1] -4.869788
```

We get a very different estimate of -4.87. This is because the Wald Estimator doesn't allow us to include the fixed effects we included above. Had we not included the fixed effects, we would have gotten a similar estimate to the Wald estimator:

```
ivreg(CIRI ~ aid | Colony, data = aid_data) |>
  modelsummary(
    statistic = "std.error",
    coef_map = c("aid" = "Effect of Aid (LATE)"),
    gof_map = c("nobs", "r.squared"),
    add_rows = tribble(
      ~term, ~CIRI,
      "Country FE", "No",
      "Year FE", "No"
    ),
    output = "markdown"
  )
```

	(1)
Effect of Aid (LATE)	-4.870 (9.476)
Num.Obs.	1792
R2	-3.445
Country FE	No
Year FE	No

i Feedback

Feedback to the problem set will be provided here.